

WASP-52b

R-1488

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_191001 · created 2026-08-01T13:35:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458757.8032 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.063
Transit depth (Rp/Rs)^2	4.67 +/- 2.74 [%]
Orbital Inclination (inc)	88.73 +/- 2.76
Airmass coefficient 1 (a1)	2.14 +/- 1.38
Airmass coefficient 2 (a2)	-1.13 +/- 1.53
Scatter in the residuals of the lightcurve fit is	64.11 %
Best Comparison Star	#3 - [77, 366]
Optimal Aperture	2.53
Optimal Annulus	3.0
Transit Duration (day)	0.074 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.063 vs 0.1646 (deviation 0.82σ)
Duration fit vs published	0.074 d vs 0.0758 d (deviation 0.23σ)
Mid-time O–C	-11.1 min
Depth SNR	3.4
Residual scatter	64.11 %

Light curve

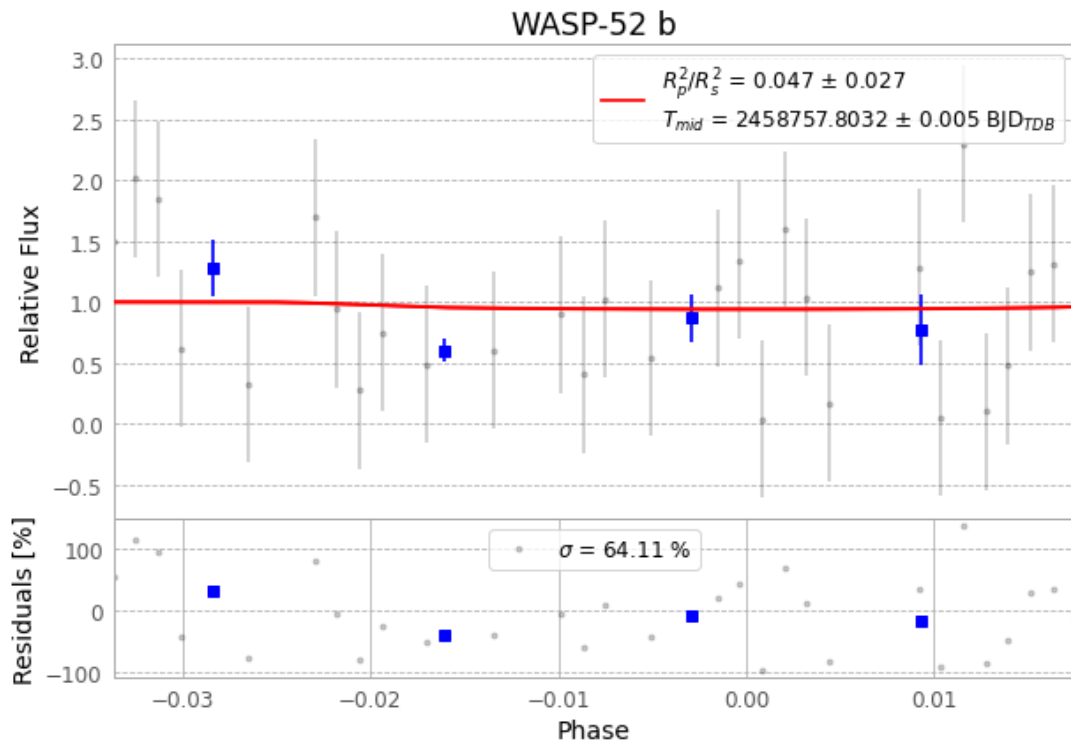


Figure 1 — FinalLightCurve_WASP-52 b_2019-09-30.png (SHA-256 20ea90bac2767a0b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [522, 154], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 83806613b25a6dd8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-52191001053612.FITS → WASP-52191001091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-191001.log (181ea6228595...), FinalLightCurve_WASP-52 b_2019-09-30.png (20ea90bac276...), FinalLightCurve_WASP-52 b_2019-09-30.csv (194f5421d051...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 13:35Z.